



EVIDENCE TO ACTION

How Lumen works

A daily automated pipeline connects public evidence, transparent scoring and direct outreach.

1 COLLECT

n8n triggers source-specific ingestion once per day.

2 STORE

PostgreSQL preserves raw readings before calculation.

3 SCORE

A tested engine normalizes available cohort signals.

4 RANK

Need versus coverage, adjusted for underfunding.

5 EXPLAIN

Dashboard shows trends, evidence and grounded briefs.

6 ACT

Priority briefs reach users through Telegram and email.

SCORING METHOD

Transparent, cohort-relative prioritization

- Log-transform inputs so very large displacement values do not flatten the cohort.
- Normalize available signals to 0–1 across crises scored in the same run.
- Estimate need from displacement, appeal size and active-disaster count.
- Compare need with media coverage, then amplify and add the funding-gap signal.

DATA SOURCES

UNHCR Displaced people

OCHA FTS Appeals and funding gap

GDELT Media coverage volume

ReliefWeb Active disaster count

ReliefWeb activates after appname approval.

GROUNDING SAFEGUARDS

- Scores store algorithm version and inputs.
- Every brief references its exact score row.
- Numeric facts are rendered deterministically.
- Groq failure triggers a grounded fallback.

No source failure or AI outage is allowed to erase the evidence trail.